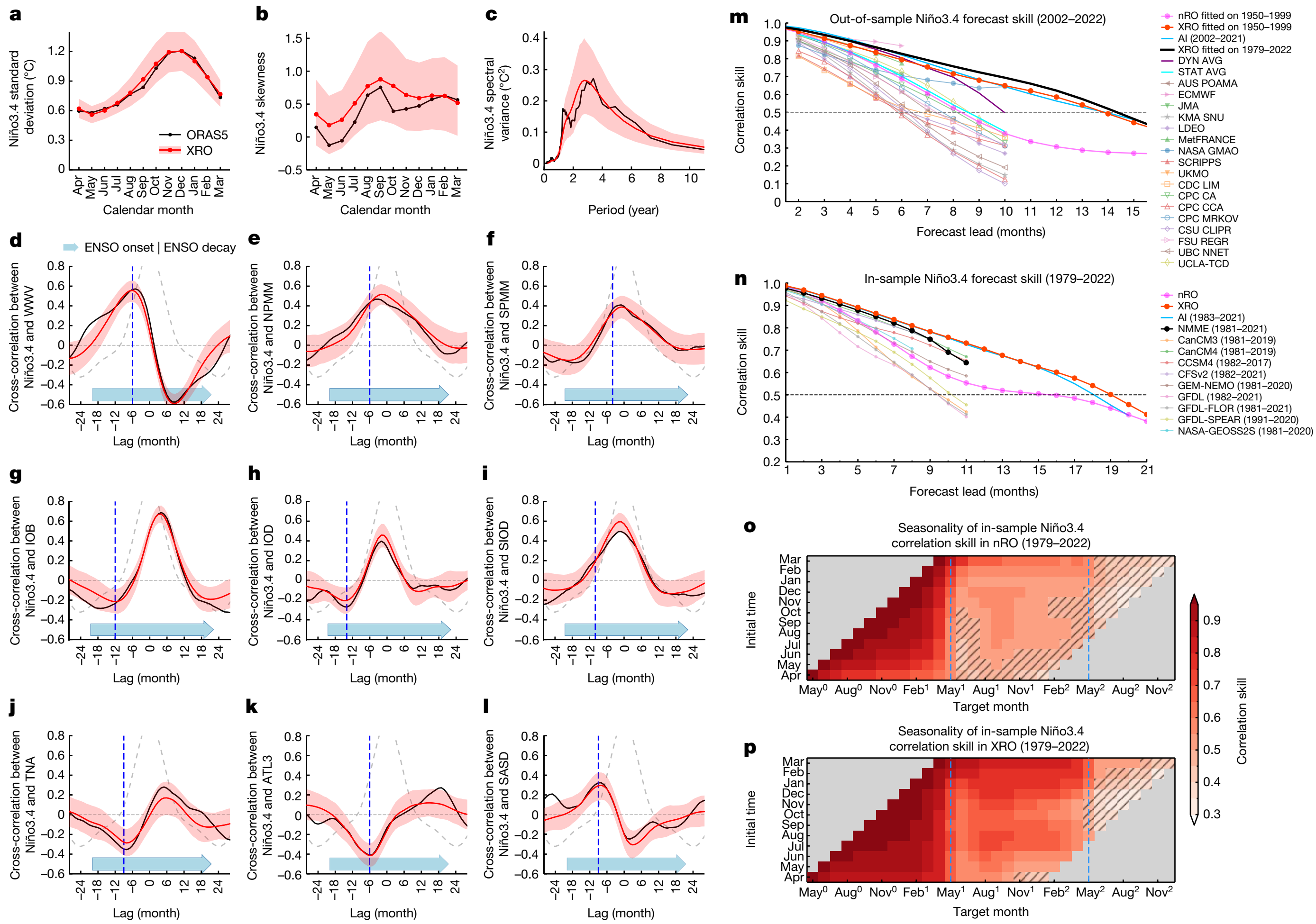

Fig. 2 | Superior efficacy of the XRO in simulating and reforecasting ENSO. **a**–**c**, Seasonally varying standard deviation (**a**), skewness (**b**) and power spectrum (**c**) of Niño3.4 using ORAS5 (black) and the XRO stochastic simulation (red). **d**–**l**, Monthly cross-correlations between Niño3.4 and different indices: WWV (**d**), NPMM (**e**), SPMM (**f**), IOB (**g**), IOD (**h**), SIOD (**i**), TNA (**j**), ATL3 (**k**) and SASD (**l**). Dashed grey curves show the autocorrelation of Niño3.4 and vertical blue dashed lines denote lead times of 6 (WWV), 6 (NPMM), 4 (SPMM), 12 (IOB), 14 (IOD), 10 (SIOD), 9 (TNA), 6 (ATL3) and 9 (SASD) months. Abscissas indicate lead time in months (negative values representing Niño3.4 lag). Red shading indicates the 10–90% ensemble spread of simulated 43 year segments, obtained from splitting a 43,000 year XRO simulation into 1,000 non-overlapping parts. **m**, All-months correlation skill of 3 month running mean Niño3.4 as a function of forecast lead for forecasts verified on 2002–2022 for the out-of-sample nRO (magenta) and XRO (red) fitted on 1950–1999, the AI model (blue), the XRO control fitted on 1979–2022 (black) and IRI operational models, the ensemble mean of dynamical models (dark purple curve) and ensemble mean of statistical models (dark cyan curve). **n**, Same as **m**, but for Niño3.4 forecast skills for nRO (magenta) and XRO (red) control forecasts, AI model forecasts (blue), and NMME dynamical model forecasts (multi-model ensemble mean in black, ensemble means from individual models in other colours). Validation period is 1979–2022 except for the AI and NMME models (period indicated in the legend). **o**,**p**, Correlation skill of nRO (**o**) and XRO (**p**) Niño3.4 forecasts as a function of initialization month (ordinate) and target month (abscissa; superscripts 0, 1 and 2 denote the current and subsequent years, respectively). Hatching highlights forecasts with a correlation less than 0.5. The dashed vertical blue lines denote the SPB. The XRO accurately simulates the fundamental observed ENSO characteristics, its lead-lag relationships with other climate modes and provides skilful forecasts at lead times up to 16–18 months.

interactions for ENSO forecast skill regardless of the intrinsic decadal changes in ENSO predictability[33,34].

To get sufficient sample sizes of ENSO events, we next focus on the satellite era (1979–2022) and perform in-sample control reforecasts using the XRO and nRO (denoted as XRO and nRO in the figures, respectively; 'Control XRO and nRO reforecasts' section in the Methods). The nRO ranks in the middle of the skill range for the existing state-of-the-art dynamical prediction systems (Fig. 2n). The XRO systematically outperforms the individual dynamical models and multi-model ensemble mean. The correlation skill of XRO is still above 0.5 at a lead time of 18 months, which is again comparable to the most skilful AI model (Fig. 2n). We also use two further approaches to confirm the robustness of the XRO parameter fitting and reforecasting performance during 1979–2022 ('Cross-validated reforecasts' and 'Large ensemble simulations and perfect model reforecasting experiments' sections in the Methods and Supplementary Fig. 8). First, the XRO cross-validated by sequentially leaving *n* year data out still provides skilful prediction of Niño3.4 SSTA up to 17 months in advance and is insensitive to the exclusion of a range between 2 and 7 years of data (Supplementary Fig. 8a). Second, the XRO was repeatedly trained using each member of large ensemble CGCM simulations (LENS) and forecasted on the same member (same-member experiment) and an independent realization (cross-member experiment), respectively. All four LENS models' perfect experiments using the same observational record length (43 years) demonstrate the uncertainty in parameter estimation leads to XRO reforecasting correlation skill error of less than 0.1 within 21 lead months (Supplementary Fig. 8b–d).

We further assess the seasonality of the Niño3.4 forecast correlation skill during 1979–2022 in Fig. 2o,p and Supplementary Fig. 9. Like most of the dynamical models, the nRO shows a pronounced spring